

DATA ANALYTICS PRACTICE SERIES



Apple Sales Analytics

Pandas Workbook

Beginner to Advanced

Project-Based Data Analytics Practice

15 Blocks • 75 Questions • Power BI Ready

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NOTE

Each block progresses from foundational recall (Easy) through applied reasoning (Medium) to a Business Challenge (Hard) rooted in the Apple Sales dataset.

PART TWO

Pandas Workbook

Fifteen blocks. Seventy-five questions. From loading raw CSV files to a fully optimized, Power BI-ready Master Sales DataFrame built on real Apple sales data.

BLOCK 01

Loading & Understanding the Dataset

TOPICS COVERED

read_csv()

head()

tail()

shape

columns

dtypes

info()

describe()

Q1

EASY

Load all five datasets (`sales.csv` , `products.csv` , `stores.csv` , `category.csv` , `warranty.csv`) into separate Pandas DataFrames.

Q2

EASY

Display the first 10 records, last 10 records, shape, columns, and data types of every dataset.

Q3

MEDIUM

Create a summary table showing the number of rows, columns, missing values, and memory usage for each dataset.

Q4

MEDIUM

Identify the Fact Table and Dimension Tables in the Apple Sales dataset and explain their purpose.

Q5

HARD – BUSINESS CHALLENGE

Prepare a professional Dataset Overview Report containing dataset name, row count, column count, primary key, foreign keys, and business purpose of each table.

BLOCK 02

Row & Column Selection

TOPICS COVERED

[]

loc[]

iloc[]

Column Selection

Row Selection

Q1

EASY

Display only the `Product Name` and `Price` columns from the Products dataset.

Q2

EASY

Display the first 20 rows and last 20 rows from the Sales dataset.

Q3

MEDIUM

Extract `Sale ID`, `Product ID`, `Store ID`, and `Quantity` using `iloc[]`.

Q4

MEDIUM

Display only stores belonging to a selected country using `loc[]`.

Q5

HARD – BUSINESS CHALLENGE

Create a new DataFrame named `sales_basic` containing only `Sale ID`, `Product ID`, `Store ID`, and `Quantity`.

BLOCK 03

Filtering & Conditional Selection

TOPICS COVERED

Boolean Filtering

isin()

between()

Multiple Conditions

Q1

EASY

Display all products having a Price greater than ₹1000.

Q2

EASY

Display all sales where Quantity is greater than 2.

Q3

MEDIUM

Display all warranty claims whose status is Approved.

Q4

MEDIUM

Display all stores located in India.

Q5

HARD – BUSINESS CHALLENGE

Prepare a filtered dataset showing Premium Apple products sold only through Indian stores with complete store and product information.

BLOCK 04

Data Cleaning

TOPICS COVERED

isnull()

fillna()

dropna()

duplicated()

rename()

Q1

EASY

Find missing values in every dataset.

Q2

EASY

Find duplicate records in the Sales dataset.

Q3

MEDIUM

Rename all column names using snake_case format.

Q4

MEDIUM

Convert every column name into lowercase and remove unnecessary spaces.

Q5

HARD – BUSINESS CHALLENGE

Create a cleaned version of the Sales dataset named `clean_sales` which is ready for further analysis.

BLOCK 05

Data Type Conversion

TOPICS COVERED

astype()

to_datetime()

Category dtype

Numeric Conversion

Q1

EASY

Convert `sale_date` into DateTime format.

Q2

EASY

Convert `launch_date` into DateTime format.

Q3

MEDIUM

Convert Country, Category, and Warranty Status into Category data type.

Q4

MEDIUM

Compare memory usage before and after data type optimization.

Q5

HARD – BUSINESS CHALLENGE

Create a final `analysis_ready_products` DataFrame with optimized data types and proper formatting for future business analysis.

BLOCK 06

String Operations

TOPICS COVERED

`str.upper()``str.lower()``str.strip()``str.replace()``str.contains()`**Q1**

EASY

Convert all Product Names into uppercase.

Q2

EASY

Remove extra spaces from all text columns wherever applicable.

Q3

MEDIUM

Find all products whose names contain the word "iPhone".

Q4

MEDIUM

Replace the word "Apple" with "AAPL" in the Product Name column.

Q5

HARD – BUSINESS CHALLENGE

Create a searchable Product Catalog containing Product Name, Category, Price, and Launch Date sorted alphabetically.

BLOCK 07

Date & Time Analysis

TOPICS COVERED

dt.year

dt.month

dt.day_name()

dt.quarter

Sorting Dates

Q1

EASY

Extract the Year from the Sale Date column.

Q2

EASY

Extract the Month Name from the Sale Date column.

Q3

MEDIUM

Extract the Quarter and Weekday from the Sale Date column.

Q4

MEDIUM

Identify the month having the highest number of sales records.

Q5

HARD – BUSINESS CHALLENGE

Prepare a Monthly Sales Trend table containing Year, Month, Number of Sales, and Total Quantity Sold.

BLOCK 08

Sorting, Ranking & Sampling

TOPICS COVERED

sort_values()

rank()

nlargest()

nsmallest()

sample()

Q1

EASY

Sort all products based on Price in descending order.

Q2

EASY

Display the Top 10 most expensive Apple products.

Q3

MEDIUM

Assign a Price Rank to every product.

Q4

MEDIUM

Take a random sample of 500 sales records for quick inspection.

Q5

HARD – BUSINESS CHALLENGE

Prepare a Premium Product Report showing Product Name, Category, Launch Date, Price, and Price Rank.

BLOCK 09

GroupBy & Aggregation

TOPICS COVERED

groupby()

agg()

sum()

mean()

count()

nunique()

Q1

EASY

Calculate the Total Quantity Sold for every Product.

Q2

EASY

Count the total number of products available in each Category.

Q3

MEDIUM

Find the total number of sales handled by every Store.

Q4

MEDIUM

Calculate the Average Quantity Sold by each Country.

Q5

HARD – BUSINESS CHALLENGE

Prepare a Store Performance Report containing Store Name, Country, Total Sales Records, Total Quantity Sold, and Average Quantity Sold.

BLOCK 10

Merge & Join

TOPICS COVERED

merge()

inner join

left join

right join

outer join

Q1

EASY

Merge the Sales dataset with the Products dataset using Product ID.

Q2

EASY

Merge Store information into the Sales dataset.

Q3

MEDIUM

Merge Category information into the Products dataset.

Q4

MEDIUM

Merge Warranty information with the Sales dataset using Sale ID.

Q5

HARD – BUSINESS CHALLENGE

Create a final Master Sales DataFrame containing Sale Date, Product Name, Category, Store Name, Country, Quantity, Price, and Warranty Status. This DataFrame will be used throughout the remaining project and later in Power BI.

BLOCK 11

Pivot Tables & Crosstab

TOPICS COVERED

pivot_table()

crosstab()

aggfunc

margins

fill_value

Q1

EASY

Create a Pivot Table showing Total Revenue by Country.

Q2

EASY

Create a Pivot Table showing Total Quantity Sold by Category.

Q3

MEDIUM

Create a Pivot Table showing Average Revenue for each Store in every Country.

Q4

MEDIUM

Create a Crosstab showing Warranty Status vs Category.

Q5

HARD – BUSINESS CHALLENGE

Create a Dashboard-Ready Pivot Table containing Country, Category, Revenue, Quantity Sold, Average Quantity, and Warranty Claims.

BLOCK 12

Apply(), Map() & Transform()

TOPICS COVERED

apply()

map()

lambda

transform()

Q1

EASY

Create a new column Revenue = Quantity × Price.

Q2

EASY

Create a Price Category column with values Budget, Mid-Range, Premium, and Luxury based on Product Price.

Q3

MEDIUM

Create a Store Performance column using Total Revenue:

- Excellent
- Good
- Average
- Poor

Q4

MEDIUM

Calculate each Store's contribution (%) to Total Revenue using `transform()`.

Q5

HARD – BUSINESS CHALLENGE

Create a final KPI DataFrame containing Store Name, Country, Revenue, Revenue Contribution (%), Average Quantity, and Performance Category.

BLOCK 13

Rolling, Expanding & Time Series Analysis

TOPICS COVERED

rolling()

expanding()

shift()

cumsum()

Q1

EASY

Calculate the Running Total Revenue based on Sale Date.

Q2

EASY

Calculate a 7-Day Moving Average of Revenue.

Q3

MEDIUM

Create a column showing the Previous Day Revenue using `shift()`.

Q4

MEDIUM

Calculate the Revenue Growth (%) compared to the previous day.

Q5

HARD – BUSINESS CHALLENGE

Build a complete Sales Trend Report containing Date, Daily Revenue, Previous Revenue, Growth (%), Running Revenue, and 7-Day Moving Average.

BLOCK 14

Memory Optimization & Performance

TOPICS COVERED

category

memory_usage()

value_counts()

MultiIndex

Q1

EASY

Convert Country, Category, Warranty Status, and Product Name into Category dtype wherever appropriate.

Q2

EASY

Compare the memory usage before and after optimization.

Q3

MEDIUM

Find the Top 10 Categories by Product Count.

Q4

MEDIUM

Create a MultiIndex using Country and Category, then generate a summary report.

Q5

HARD – BUSINESS CHALLENGE

Optimize the Master Sales DataFrame for maximum performance while preserving all business information and document every optimization applied.

BLOCK 15

Final Business Analysis Project

TOPICS COVERED

Business Analysis

KPIs

Executive Reporting

Dashboard Dataset Preparation

Q1

EASY

Identify the Top 10 Products based on Total Revenue.

Q2

EASY

Identify the Top 10 Stores based on Total Revenue.

Q3

MEDIUM

Create a Country-wise Business Performance Report containing Revenue, Quantity Sold, Average Revenue per Sale, and Warranty Claim Rate.

Q4

MEDIUM

Perform a complete Category Performance Analysis including Revenue, Quantity, Product Count, and Warranty Claims, then rank all categories from best to worst.

Q5

HARD – BUSINESS CHALLENGE

Prepare the Final Business Dataset for Power BI by creating a clean, analysis-ready DataFrame containing:

- Sale ID
- Sale Date
- Product Name
- Category
- Store Name
- Country
- Quantity
- Price
- Revenue
- Warranty Status
- Month
- Quarter
- Year
- Store Performance
- Price Category

This dataset should require no additional cleaning before importing into Power BI.



Pandas Workbook Complete

- ✓ 15 Blocks
- ✓ 75 Questions
- ✓ Apple Sales Dataset
- ✓ Beginner → Advanced
- ✓ Business-Oriented
- ✓ GitHub Project Ready
- ✓ Power BI Ready
- ✓ Interview Ready